

B. Tech Degree VI Semester Examination April 2011

ME 602 DYNAMICS OF MACHINERY (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART – A (Answer ALL questions)

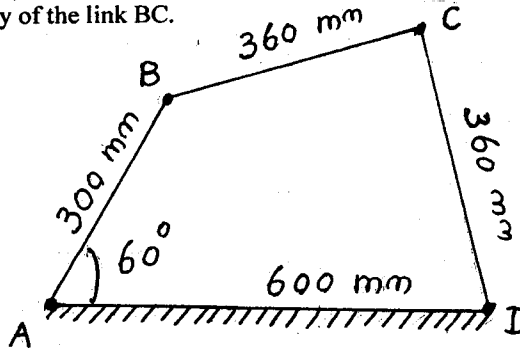
(8 x 5 = 40)

- I. (a) State and explain D'Alembert's principle.
 (b) Give an account of the two inversions of a slider-crank mechanism.
 (c) What is coefficient of fluctuations of speed? Obtain the expression for the coefficient of fluctuation of speed in terms of maximum fluctuation of energy and kinetic energy of the flywheel at mean speed.
 (d) Discuss the effect of gyroscopic couple on an aeroplane.
 (e) Explain static balancing and dynamic balancing.
 (f) What do you mean by balancing machines? Describe any one type of balancing machine.
 (g) Derive the relation for ratio of belt tensions in a flat belt drive.
 (h) With the help of a neat sketch explain belt transmission dynamometer.

PART – B

(4 x 15 = 60)

- II. (a) In a pin jointed four bar mechanism as shown in the figure, $AB = 300$ mm, $BC = CD = 360$ mm and $AD = 600$ mm. The angle $BAD = 60^\circ$. The crank AB rotates uniformly at 100 rpm. Locate all the instantaneous centers and find the angular velocity of the link BC .



- (b) Explain shaking forces and shaking moments.

OR

- III. A horizontal steam engine running at 240 rpm has a bore of 20 cm and stroke of 36 cm. The piston rod is 2 cm in diameter and connecting rod length is 90 cm. The mass of the reciprocating parts is 7 Kg and the frictional resistance is equivalent to a force of 500 N. Determine the following when the crank is at 120° from the inner dead center, the mean pressure being 5×10^2 N/m² on the cover side and 1×10^2 N/m² on crank side.

- (i) thrust on the connecting rod
 (ii) thrust on the cylinder walls
 (iii) load on the bearings
 (iv) turning moment on the crank shaft.

(15)

(P.T.O)

- IV. The intercepted areas between the output torque curve and the mean resistance line of a turning moment diagram for a multicylinder engine, taken in order from one end are as follows:

– 0.35, + 4.10, – 2.85, + 3.25,
– 3.35, + 2.60, – 3.65, + 2.85, – 2.60 sq.cm.

The diagram has been drawn to a scale of 1 cm = 700 N-m and 1 cm = 45° . The engine speed is 900 rpm and the fluctuation in speed is not to exceed 2% of the mean speed.

Find a suitable diameter and cross section of the fly wheel rim if the safe centrifugal stress is limited to $7 \times 10^6 \text{ N/m}^2$. The density of the material of the flywheel may be taken as 7200 Kg/m^3 . The rim is rectangular with the width 2 times the thickness. Neglect effect of arms etc. (15)

OR

- V. Each ball of a Porter governor has a mass of 3 Kg and the mass of the sleeve is 15 Kg. The governor has equal arms each of 200 mm length and pivoted on the axis of rotation. When the radius of rotation of the ball is 120mm, the sleeve begins to rise up 160 mm at the maximum speed. Determine the

- (i) range of speed
- (ii) lift of the sleeve
- (iii) effort of the governor
- (iv) power of the governor.

What will be the effect of friction at the sleeve if it is equivalent to 8N? (15)

- VI. A shaft supported in bearings that are 1.6m apart projects 400mm beyond bearings at each end. It carries three pulleys one at each end and one at the center of its length. The masses of the end pulleys are 40 Kg and 22Kg and their centers of mass are at 12mm and 18 mm respectively from the shaft axes. The mass of the center pulley is 38 Kg and its center of mass 15 mm from the shaft axis. The pulleys are arranged in a manner that they give static balance. Determine

- (i) the relative angular positions of the pulleys
- (ii) the dynamic forces developed on the bearings when the shaft rotates at 210 rpm. (15)

OR

- VII. A four crank engine has the two outer cranks set at 120° to each other and their reciprocating masses are each 400 Kg. The distance between the planes of rotation of adjacent cranks are 45, 75 and 60 cm. If the engine is to be in complete primary balance, find the reciprocating mass and the relative angular positions for each of the inner cranks.

If the length of each crank be 30 cm, the length of each connecting rod 10cm and the speed of rotation 240 rpm. What is the maximum secondary unbalanced force? (15)

- VIII. Determine the width of a 9.75 mm thick leather belt required to transmit 15 KW from a motor running at 900 rpm. Diameter of the driving pulley of the motor is 30 cm. The driven pulley runs at 300 rpm and the distance between the center of the two pulleys is 3 meters. The mass of the leather is $1 \times 10^{-4} \text{ Kg/cm}^3$. Maximum allowable stress in the leather is 250 N/cm^2 . Coefficient of friction between the leather and pulley is 0.30. Assume open belt drive and neglect the sag and slip of the belt. (15)

OR

- IX. (a) Explain the working of a shoe brake. (5)
(b) A band brake acts on the $3/4$ of circumference of a drum of 450 mm diameter which is keyed to the shaft. The band brake provides a braking torque of 225 N-m. One end of the band is attached to a fulcrum pin of the lever and the other end to a pin 100 mm from the fulcrum. If the operating force is applied at 500 mm from the fulcrum and the coefficient of friction is 0.25, find the operating force when the drum rotates in the

- (i) anticlockwise direction
- (ii) clockwise direction. (10)